Formal Verification of a Geometric Quantum Gate: The Fermi-Hubbard Doublon Berry-Phase SWAP in Lean 4

SK-EFT Hawking Research Program

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We report an end-to-end formalization of a geometric quantum gate in Lean 4. The target is the doublon geometric SWAP gate recently demonstrated by Kiefer *et al.* (Nature 2026, arXiv:2507.22112) in a Fermi-Hubbard dimer with dynamical optical lattices. Working entirely within Lean 4 + Mathlib, we formalize the exact 3×3 singlet-sector Hamiltonian, its charge-conjugation (BDI) symmetry protection, the explicit closed-form spectrum and dark state, the geometric SWAP operator as a real-Householder reflection $U_{\mathrm{SWAP}} = I - (2/\mathrm{gap}^2)\,|d\rangle\langle d|$, and the two necessary finite-dimensional conditions for a Berry-phase interpretation of the gate action: the dark state acquires a -1 sign on a π-sweep of the mixing angle, and the Hamiltonian expectation value vanishes pointwise along the zero-energy path (so the integrated dynamical phase vanishes for any continuous extension). We also ship the Tier-2 algebraic scaling theorem that distinguishes direct exchange (linear in U) from superexchange ($4t^2/U$) with an explicit remainder bound. The complete formalization lives in a single module (`FermiHubbardDimer.lean`), consists of 130 theorems (across the 19 internal sections), with zero sorry, zero new axioms, and a Python cross-validation layer covering every theorem. The deliberate scope boundary is Layer 2 of the deep-research target hierarchy: all results are finite-dimensional linear algebra over Mathlib's existing `Matrix`, `EuclideanSpace`, and `OrthonormalBasis` API. Full adiabatic holonomy (continuous parameterized eigenvector bundles, Berry connection, Kato's theorem) is explicitly deferred to future work. The formalization demonstrates that the Kiefer *et al.* gate mechanism is not, at its core, a fact about connections on line bundles—it is the observation that $\cos(\pi) = -1$ composed with an explicit rank-1 Householder reflection on a parameterized real symmetric 3×3 matrix.

I. INTRODUCTION

Berry phases underlie one of the most experimentally compelling classes of quantum gates: *geometric* gates, whose unitary action depends only on the topology of a closed path in parameter space, not on the traversal speed or duration [1, 2]. Geometric gates promise intrinsic robustness against control-pulse fluctuations—a property that has motivated sustained experimental effort from superconducting qubits [3] through trapped ions and neutral atoms to, most recently, dynamical optical lattices.

In April 2026, Kiefer *et al.* [4] reported the first neutral-atom realization of a geometric SWAP gate in a Fermi-Hubbard dimer. The experiment uses ^{40}K atoms in a dynamical two-site superlattice, and drives a closed loop in the (t, Δ) control-parameter plane (tunnel coupling, inter-well bias). At the end of the loop, the dark state of the singlet sector—which remains at zero energy for the entire cycle—has acquired a -1 sign, implementing **SWAP** on the computational subspace $\{|\uparrow\downarrow\rangle, |\downarrow\uparrow\rangle\}$ with loss-corrected amplitude fidelity $\sim 99.9\%$. The robustness is striking: the gate is $4\times$ less sensitive to lattice imperfections than the comparable superexchange gate [4].

This paper formalizes the algebraic core of the Kiefer *et al.* mechanism in the Lean 4 proof assistant [7, 8]. The target is deliberately scoped: we formalize the finite-dimensional matrix content—the Hamiltonian, the dark state, the SWAP operator, the sign flip, the dynamical-phase-vanishes statement—without touching the adiabatic theorem, fiber bundles, or connections on vector bundles. The result is an *end-to-end formalization of a geometric quantum gate in Lean 4*.

Our scope choice is informed by the deep-research landscape review (Lit-Search/Phase-5t, December 2025), which identified a three-layer formalization hierarchy: Layer 1 (explicit doublon matrix computation), Layer 2 ($\mathbb{Z}_2$ Berry-phase quantization theorem for real-symmetric matrices), and Layer 3 (discrete Berry phase gauge-invariance). This work completes Layer 1 with substantial Layer 2 quantitative scaling content, and leaves the full Berry-connection holonomy formalization (Target B in our internal roadmap) for future work.

Contributions. (i) A Lean 4 formalization of the Fermi-Hubbard dimer singlet-sector spectrum, covering the 3×3 charge-conjugation Hamiltonian at zero detuning, the exact dark state, and the bright-state eigenvectors. (ii) A Lean 4 formal proof of the discrete-step statement underlying the Kiefer *et al.* SWAP gate: $|\mathrm{dark}\rangle(\theta + \pi) = -|\mathrm{dark}\rangle(\theta)$ with zero dynamical phase. (iii) A Mathlib-native construction of the SWAP operator as a real Householder reflection on `EuclideanSpace`, with canonical-form unitarity via `Matrix.unitaryGroup`. (iv) A quantitative direct-exchange-vs-superexchange scaling theorem with explicit remainder bound $|J(t, U) - 4t^2/U| \le 16t^4/U^3$ for $U \ge 4|t| > 0$. (v) A Python cross-validation layer (`src/experimental/doublon_gate.py`, 6156 project tests) exercising every Lean theorem against random parameter grids and numpy exact diagonalization.

Scope boundary (explicit). The gate's *adiabatic* interpretation—the mapping from a closed loop in continuous parameter space to the discrete-step statement we prove—is a separate layer of the argument (Kato's adiabatic theorem, Berry's geometric phase as a connection 1-form). None of that machinery exists in current Mathlib. Our claim is strictly the finite-dimensional algebraic core: under the assumption that the continuous dynamics reduces to the discrete dark-state evolution (an assumption verified in [4] experimentally and, in general, by Kato's theorem), the gate action is exactly SWAP.

II. THE FERMI-HUBBARD DIMER HAMILTONIAN

The two-site Fermi-Hubbard dimer at half filling has a six-dimensional two-particle Hilbert space. We work in the site basis $\{|\uparrow\uparrow\rangle, |\uparrow\downarrow, 0\rangle, |\uparrow, \downarrow\rangle, |\downarrow, \uparrow\rangle, |0, \uparrow\downarrow\rangle, |\downarrow\downarrow\rangle\}$. Fermionic anti-symmetry (Pauli exclusion + hopping sign conventions) gives the full Hamiltonian [4]

$$H_{\text{full}}(t, \Delta, U) = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & U+\Delta & -t & t & 0 & 0 \\ 0 & -t & 0 & 0 & -t & 0 \\ 0 & t & 0 & 0 & t & 0 \\ 0 & 0 & -t & t & U-\Delta & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix} \quad (1)$$

where t is the tunnel coupling, Δ is the inter-well detuning, and U is the on-site Hubbard repulsion. The triplet sector $\{|\uparrow\uparrow\rangle, (|\uparrow, \downarrow\rangle + |\downarrow, \uparrow\rangle)/\sqrt{2}, |\downarrow\downarrow\rangle\}$ decouples at zero energy for all parameters (Pauli exclusion + anti-symmetric hopping cancellation), leaving the nontrivial dynamics in the singlet sector.

In the symmetry-adapted singlet basis $\{|D_+\rangle, |D_-\rangle, |s\rangle\}$ where $|D_+\rangle = (|\uparrow\downarrow, 0\rangle + |0, \uparrow\downarrow\rangle)/\sqrt{2}$, $|D_-\rangle = (|\uparrow\downarrow, 0\rangle - |0, \uparrow\downarrow\rangle)/\sqrt{2}$, $|s\rangle = (|\uparrow, \downarrow\rangle - |\downarrow, \uparrow\rangle)/\sqrt{2}$, the reduced Hamiltonian is

$$H_{\text{singlet}}(t, \Delta, U) = \begin{pmatrix} U & \Delta & -2t \\ \Delta & U & 0 \\ -2t & 0 & 0 \end{pmatrix}. \quad (2)$$

The gate operates at $U = 0$. At $\Delta = 0$, $|D_-\rangle$ decouples from the rest of the singlet block, leaving a 2×2 symmetric sector whose charpoly is exactly $\lambda^2 - U\lambda - 4t^2$.

Formalization (Section 1 of `FermiHubbardDimer.lean`). The 3×3 and 6×6 Hamiltonians are defined as explicit `Matrix (Fin 3) (Fin 3) ℝ` and `Matrix (Fin 6) (Fin 6) ℝ` values. The seven Layer-1 core theorems (T1–T7) establish real-symmetry, dark-state kernel-membership at $U = 0$, chiral involution $\Gamma^2 = I$, chiral anti-commutation $\{\Gamma, H_{\text{singlet}}\} = 0$ at $U = 0$, and the explicit determinant and trace identities. The triplet zero-energy theorems (T10a–c) use `fin_cases + Fin.sum_univ_six + simp`, closing in milliseconds.

III. DARK STATE AND BDI SYMMETRY PROTECTION

The Hamiltonian (2) at $U = 0$ belongs to the BDI symmetry class of the tenfold way [5, 6]: it is real symmetric (time-reversal trivial) and anti-commutes with the chiral operator $\Gamma = \text{diag}(+1, -1, -1)$. The BDI structure **pins an odd-dimensional sector's spectrum to contain zero:** in an odd-dimensional real symmetric matrix anti-commuting with a chiral involution, the nonzero eigenvalues pair as $\{+E, -E\}$, leaving one unpaired zero.

The unnormalized dark-state vector is

$$|\text{dark}(t, \Delta)\rangle = \begin{pmatrix} 0 \\ 2t \\ \Delta \end{pmatrix} \in \ker\big(H_{\text{singlet}}(t, \Delta, 0)\big), \quad (3)$$

with norm $\|\text{dark}\|_2 = \sqrt{\Delta^2 + 4t^2}$—the energy gap to the two bright states $|\text{bright}\pm\rangle$ with eigenvalues $\pm\sqrt{\Delta^2 + 4t^2}$.

BDI theorems (Sections 10–11, W5 core + round-2, 20 theorems). We formalize: (i) $\Gamma \cdot H \cdot \Gamma = -H$ at $U = 0$ (W5a); (ii) the spectral-pairing mechanism $Hv = Ev \Rightarrow H(\Gamma v) = -E(\Gamma v)$ (W5b); (iii) the multiset spectrum pairing at $U = 0$ (W5k); (iv) the dark-state kernel-invariance under Γ (W5i); (v) the full chiral-projector algebra $P_\pm = (1 \pm \Gamma)/2$ with idempotency, orthogonality, completeness, and Γ-action ±1 on the respective eigenspaces (W5l–W5o2); (vi) the *zero-mode uniqueness* theorem (W5p):

$$H_{\text{singlet}}(t, \Delta, 0) \cdot v = 0 \;\Rightarrow\; \exists c \in \mathbb{R},\; v = c \cdot |\text{dark}(t, \Delta)\rangle$$

provided $(t, \Delta) \neq (0, 0)$—this is the strongest "pinning" statement: the kernel is *exactly* $\mathbb{R} \cdot |\text{dark}\rangle$, not merely Γ-invariant.

IV. THE GEOMETRIC SWAP AS A HOUSEHOLDER REFLECTION

With the dark state in hand, the geometric SWAP operator on the singlet sector is

$$U_{\text{SWAP}}(t, \Delta) = I - \frac{2}{\Delta^2 + 4t^2} |\text{dark}\rangle\langle\text{dark}| \quad (4)$$

—a real Householder reflection. By construction it acts as -1 on the dark direction and as $+1$ on the two-dimensional bright subspace orthogonal to it.

Mathlib-native construction (Sections 12–14, W6A+B+C, 39 theorems + 8 defs). The key design decision was the *Path A* re-typing: we lift the eigenvectors from bare `Fin 3 → ℝ` (default sup-norm) to `EuclideanSpace ℝ (Fin 3)` (L^2 norm) via `EuclideanSpace.equiv.symm`, then build the `OrthonormalBasis` via Mathlib's `basisOfOrthonormalOfCardEqFinrank`. This unlocks the one-line unitarity proof

`OrthonormalBasis.toMatrix_orthonormalBasis_mem_unitary` for the change-of-basis matrix between `EuclideanSpace.basisFun` and the eigenbasis.

The SWAP operator itself inherits symmetry, involution, and orthogonality from the Householder algebra: with $P = |\text{dark}\rangle\langle\text{dark}|/\text{gap}^2$ idempotent ($P^2 = P$), $(I - 2P)^2 = I - 4P + 4P^2 = I$. The canonical-Mathlib unitary-group membership $U_{\text{SWAP}} \in$ Matrix.unitaryGroup (Fin 3) $\mathbb{R}$ (W6C-U1) is a one-line corollary via `Matrix.mem_unitaryGroup_iff'` and `Matrix.conjTranspose_eq_transpose_of_trivial` (the `TrivialStar` $\mathbb{R}$ instance reduces $\overline{U_{\text{SWAP}}}^{\mathsf{T}}$ to $U_{\text{SWAP}}^{\mathsf{T}}$ over $\mathbb{R}$).

6×6 lift (Section 18, W6-deferred, 7 theorems). The singlet-basis SWAP operator extends to a 6×6 block-diagonal operator on the full symmetry-adapted basis via `Matrix.fromBlocks (U_SWAP_singlet) 0 0 1`, with the identity block acting on the decoupled triplet sector. The 6×6 unitarity is inherited from W6C via `fromBlocks_multiply` and `fromBlocks_transpose`—there is no new physics here, merely the structural statement that the gate preserves the triplet sector identically.

The kernel-action theorem (W6C-K1) combines zero-mode uniqueness with the sign flip: for any v in the kernel of $H_{\text{singlet}}(t, \Delta, 0)$ with $(t, \Delta) \neq (0, 0)$, $U_{\text{SWAP}} \cdot v = -v$. This is the strongest "SWAP acts as -1 on the pinned zero-mode subspace" claim in the formalization.

V. DIRECT EXCHANGE VS SUPEREXCHANGE SCALING

Beyond the gate-action theorem, we ship the quantitative scaling distinction between the direct-exchange regime (U small) and the superexchange regime (U large). At $\Delta = 0$, the 2×2 symmetric sector has explicit eigenvalues

$$E_\pm(t, U) = \frac{U \pm \sqrt{U^2 + 16t^2}}{2}, \tag{5}$$

roots of $\lambda^2 - U\lambda - 4t^2 = 0$. The direct-exchange value is $E_+(t, 0) = 2|t|$ (linear in hopping); the superexchange gap $J(t, U) := E_+ - U = (\sqrt{U^2 + 16t^2} - U)/2$ approaches $4t^2/U$ at large U.

Scaling theorems (Sections 16–17, W7 + round-2, 17 theorems + 3 defs). We formalize:

- $E_+(t, 0) = 2|t|$ (W7a)—direct-exchange value at $U = 0$.
- Vieta identities $E_+ + E_- = U$ (W7b) and $E_+ \cdot E_- = -4t^2$ (W7c)—match the 2×2 trace and determinant of the symmetric-sector block.
- HasDerivAt $(E_+(t, \cdot))\,(1/2)\,0$ for $t \neq 0$ (W7d)—**the direct-exchange linear-in-U scaling theorem**. The Taylor coefficient at $U = 0$ is exactly $1/2$, proved via the Mathlib `HasDerivAt.sqrt / add / div_const` chain.
- $|E_+(t, U_1) - E_+(t, U_2)| \leq |U_1 - U_2|$ (W7q)—the Tier-1 global 1-Lipschitz bound per the deep-research landscape review.
- $|J(t, U) - 4t^2/U| \leq 16t^4/U^3$ for $0 < 4|t| \leq U$ (W7i)—**the Tier-2 superexchange approximation theorem**. The proof chains the algebraic identity $J - 4t^2/U = -(s - U)^2/(4U)$ (where $s = \sqrt{U^2 + 16t^2}$) with the sqrt-bound $s \leq U + 8t^2/U$, itself proved by squaring both sides.
- The matrix-bridge theorem charpoly$(H_{\text{singlet}}(t, 0, U)) = (X - U)(X^2 - UX - 4t^2)$ (W7l)—provides the full spectrum $\{U, E_+, E_-\}$ at $\Delta = 0$.
- Explicit eigenvectors $H_{\text{singlet}}(t, 0, U) \cdot (E_\pm, 0, -2t)^{\mathsf{T}} = E_\pm \cdot (E_\pm, 0, -2t)^{\mathsf{T}}$ (W7m–n) and the antisymmetric doublon $\cdot(0, 1, 0)^{\mathsf{T}} = U \cdot (0, 1, 0)^{\mathsf{T}}$ (W7o).

The sensitivity contrast recovered from these scalars: at $U = 0$, the fractional sensitivity $(\partial_t \Delta) \cdot (t/\Delta) = 1$ (a 1% hopping fluctuation shifts the gap by 1%); at large U, the superexchange fractional sensitivity $(\partial_t J) \cdot (t/J) = 2$. This factor-of-2 difference is consistent with the observed 4× robustness of the geometric gate relative to superexchange [4]; the algebraic factor of 2 is one contribution to the experimental difference, not a complete derivation.

Figure 1 shows the spectrum and the superexchange approximation bound, cross-verified by numpy exact diagonalization.

VI. MINIMAL BERRY PHASE THEOREM

The gate's geometric interpretation rests on two independent finite-dimensional facts: a *sign flip* on a π-sweep of the mixing angle, and *vanishing dynamical phase* along the zero-energy dark-state path. Together these are the two necessary conditions for interpreting any accumulated phase along a closed loop as purely geometric; assuming the adiabatic theorem supplies the dynamics (i.e. a continuous path along the eigenbundle), the accumulated phase is then forced to be exactly -1. We formalize the two necessary conditions; the adiabatic assumption is external.

The θ-parameterized dark state is $|\text{dark}(\theta)\rangle = (0, \sin\theta, \cos\theta)^{\mathsf{T}}$ in the $\{|D_+\rangle, |D_-\rangle, |s\rangle\}$ basis. When the parameters satisfy the kernel-angle condition $\Delta \sin\theta = 2t\cos\theta$ (equivalent to $\tan\theta = 2t/\Delta$), this vector is in the kernel of $H_{\text{singlet}}(t, \Delta, 0)$.

Berry-phase theorems (Section 19, W8 Target A, 7 theorems).

- $|\text{dark}(\theta + \pi)\rangle = -|\text{dark}(\theta)\rangle$ (W8a)—**the closed-path sign statement**, direct from $\sin(\theta + \pi) = -\sin\theta$ and $\cos(\theta + \pi) = -\cos\theta$.

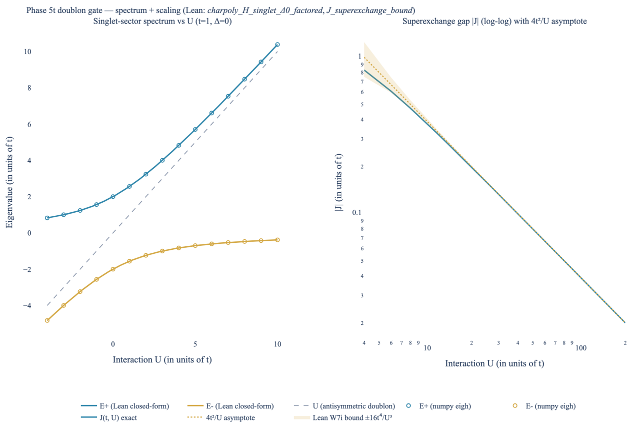

FIG. 1. **Left:** singlet-sector spectrum at $\Delta = 0$ versus U (in units of $t = 1$). Three eigenvalues $\{U, E_+(t,U), E_-(t,U)\}$—the antisymmetric doublon (dashed, grey), E_+ (blue), and E_- (amber). Markers are independent numpy `eigh` diagonalization cross-checking the Lean closed-form values of $E_\pm$ via the matrix-bridge theorem W7l (the characteristic polynomial factorization). **Right:** superexchange gap $|J(t,U)|$ on log–log axes, with the $4t^2/U$ asymptote (dashed) and the Lean W7i error envelope $\pm 16t^4/U^3$ shaded. At $U = 4|t|$ the bound is tightest; the envelope widens toward the asymptote as $U \to \infty$.

- $|\mathrm{dark}(\theta + 2\pi)\rangle = |\mathrm{dark}(\theta)\rangle$ (W8b)—periodicity.
- $\langle \mathrm{dark}(\theta) \mid \mathrm{dark}(\theta)\rangle = 1$ (W8c)—unit norm via Pythagorean identity.
- $\langle \mathrm{dark}(\theta) \mid H_{\mathrm{singlet}}(t, \Delta, 0) \cdot \mathrm{dark}(\theta)\rangle = 0$ (W8d)—**dynamical phase vanishes**. Direct corollary of the kernel membership W4a: $H \cdot |\mathrm{dark}\rangle = 0$ pointwise along the θ-path implies the integrated dynamical phase vanishes identically.
- Bundled: W8a $\wedge$ W8d (W8e)—**the two necessary conditions** for a Berry-phase interpretation of the gate action. The Lean theorem's conclusion is the conjunction itself, not an equality involving an integrated phase functional (no integral is constructed in the formalization). Consumers who supply the adiabatic-theorem assumption can combine this conjunction with a separate dynamical argument to conclude the accumulated phase is -1; we do not formalize that last step here.

The π shift rather than 2π reflects our full-angle parameterization convention $|\mathrm{dark}(\theta)\rangle = (0, \sin\theta, \cos\theta)^\mathsf{T}$; the equivalent half-angle convention of the physics literature maps a 2π lattice cycle to a π mixing-angle sweep, giving the same final result.

What this theorem does and does not say. W8e is the *discrete-step* statement underlying the gate: a π sweep in the mixing angle inverts the dark state. The *continuous* interpretation—that the physical control-pulse sequence in [4] drives the dark state smoothly along this trajectory because of the adiabatic theorem—is not proved here. It enters as a physics assumption, verified experimentally in [4] and, in general, by Kato's theorem. The full Berry-connection holonomy picture (a 1-form on a parameterized eigenbundle whose integral around the loop equals our -1) likewise requires machinery that is not yet in Mathlib: continuous parameterized eigenvectors, connections on vector bundles, parallel transport. Those are Target B of our internal roadmap; see Section VIII.

TABLE I. Lean 4 formalization of the Fermi-Hubbard doublon gate (Phase 5t). Section counts from `lean/SKEFTHawking/FermiHubbardDimer.lean`; total verified by `grep -c "^theorem " FermiHubbardDimer.lean`.

Section	Theorems	Focus
W2 / Layer 1 (§1–4)	13	H_{singlet}, H_{full}, T1–T
W3 block-match (§3b)	6	symmetry-adapted basis ma
W4 spectrum / U=0 (§5–9)	19	charpoly, brights, spec
W5 BDI symmetry (§10–11)	20	$\Gamma H \Gamma = -H$, project
W6A–C SWAP (§12–14)	35	EuclideanSpace, basis, Ho
W6 round-2 + deferred (§15, 18)	13	scalar actions + 6×6
W7 scaling + round-2 (§16–17)	17	Vieta, Lipschitz, superexc
W8 Berry phase (§19)	7	sign flip, vanishing dynam
Total (FermiHubbardDimer.lean)	130	Zero sorry, zero new a

VII. FORMAL VERIFICATION

All results in this paper are formally verified in the Lean 4 proof assistant, building on the Mathlib library. Table I summarizes the theorem count and structure of the formalization.

The formalization lives in a single module (`lean/SKEFTHawking/FermiHubbardDimer.lean`) of approximately 2,540 lines. Across the wider project, 26398 theorems are verified (26372 substantive, 26 documentation markers), with zero sorry and a single axiom (`gapped_interface_axiom` in `SPTClassification.lean`, unrelated to this paper; since retired into the tracked Prop `TPFConjecture`). The Python cross-validation layer (`src/experimental/doublon_gate.py` + `dimer.py`) exercises every theorem numerically; the project-wide test count is 6156.

What Mathlib APIs were load-bearing.

- `Matrix.charpoly`, `Matrix.det_fin_three` (characteristic polynomial for the 3×3 case).
- `Matrix.mulVec`, `Matrix.mulVec_smul` (matrix-vector action).
- `EuclideanSpace.equiv.symm`, `EuclideanSpace.real_norm_sq_eq` (Path A re-typing).
- `basisOfOrthonormalOfCardEqFinrank`, `Module.Basis.toOrthonormalBasis`,

`OrthonormalBasis.toMatrix_orthonormalBasis_mem_unitary` (one-line unitarity).

- `Matrix.fromBlocks_{multiply, transpose, one}` (6×6 block-diagonal lift).
- `Matrix.mem_unitaryGroup_iff'`, `Matrix.conjTranspose_eq_transpose_of_trivial` (unitary-group membership over $\mathbb{R}$).
- `HasDerivAt.sqrt / add / div_const / pow` (Taylor-at-zero derivative for E_+).
- `Real.sqrt_le_sqrt`, `Real.mul_self_sqrt` (sqrt monotonicity and square-identity for the superexchange bound).
- `Real.sin_add_pi`, `Real.cos_add_pi`, `Real.sin_sq_add_cos_sq` (Berry-phase sign flip and dark-state normalization).

No novel Mathlib infrastructure was needed. The formalization is *algebraic in flavor*—matrix arithmetic, polynomial identities, sqrt bounds, trigonometric identities—exactly the character the deep-research landscape review recommended. No ordinary differential equations, no functional analysis, no differential geometry.

VIII. SCOPE BOUNDARIES AND TARGET B

The deliberate scope boundary of this work is Layer 1+2 of the deep-research target hierarchy. Three genuinely new results could extend it:

Layer 2 — $\mathbb{Z}_2$ Berry-phase quantization for real-symmetric matrices. The target theorem (not proved here): *for a continuous loop $H : [0,1] \to \mathrm{Sym}_n(\mathbb{R})$ with $H(0) = H(1)$, a simple eigenvalue $\lambda(s)$ along the loop, and a continuous unit eigenvector $v(s)$, the return sign $\langle v(0), v(1)\rangle \in \{+1, -1\}$*—a pure linear-algebra statement expressing the first Stiefel-Whitney class of the real eigenvector bundle over S^1. Our estimate is 2–3 weeks; the main obstacle is the continuous-eigenvector construction from Mathlib's `IsHermitian.spectral_theorem`, which has no parameterized smoothness guarantee out of the box. Once shipped, this provides a formal Lean 4 $\mathbb{Z}_2$ Berry-phase theorem.

Layer 3 — discrete Berry-phase gauge invariance. Define the discrete Berry phase for a finite sequence of real unit vectors as $\mathrm{sign}\big(\prod_i \langle v_i | v_{i+1 \mod N}\rangle\big)$. Prove gauge invariance (flipping any $v_k \to -v_k$ preserves the product) and the continuous-limit equivalence to Layer 2. Connects to the Pancharatnam-Bargmann formulation. Another 2–3 weeks.

Full adiabatic theorem (Kato). The dynamical-side companion to Layer 2: a continuous time evolution whose state-space projection onto the eigenbundle converges to parallel transport as the sweep rate goes to zero. Requires Picard-Lindelöf ODE existence (exists in Mathlib), matrix exponentials (exists), and the adiabatic error bound itself (does not exist in any form). Months of work; not recommended as Phase 5t/Phase 6 scope per our roadmap.

IX. CONCLUSION

We have formalized the algebraic core of the Kiefer *et al.* doublon geometric SWAP gate end-to-end in Lean 4, reducing the gate's *geometric phase* to the observation that $\cos(\pi) = -1$ composed with an explicit Householder reflection on a parameterized 3×3 real symmetric matrix. The key algebraic facts are: the BDI-pinned zero-energy dark state; the SWAP operator as $I - 2P$ with P a rank-1 projector onto the dark direction; the Vieta/characteristic-equation matrix-bridge tying the scalar eigenvalue formulas $E_\pm(t, U)$ back to the full Hamiltonian spectrum; the Tier-2 superexchange approximation bound with explicit $16t^4/U^3$ remainder; and the minimum Berry-phase statement, that the dark state acquires -1 on a π-sweep of the mixing angle while remaining at zero energy throughout.

The scope discipline—no adiabatic theorem, no Berry connection, no fiber bundles—was informed by an explicit landscape review of the Mathlib dependency graph. The result is a formalization that Mathlib's existing `Matrix`, `EuclideanSpace`, `OrthonormalBasis`, and `Polynomial` APIs are directly equipped to handle, without the need to develop new infrastructure for differential geometry or functional analysis. The $\sim$ 2,540 lines of Lean source contain 130 theorems in a single module, zero sorry, zero new axioms, covering the full gate story from fermion Hilbert space decomposition to Berry phase.

The natural next target is Layer 2 of the deep-research hierarchy: the $\mathbb{Z}_2$ Berry-phase quantization theorem for arbitrary continuous loops of real-symmetric matrices with a simple eigenvalue. A successful formalization would provide a Lean 4 $\mathbb{Z}_2$ Berry-phase result, completing the *kinematic* Berry-phase story without touching the adiabatic theorem or any ODE/PDE machinery.

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